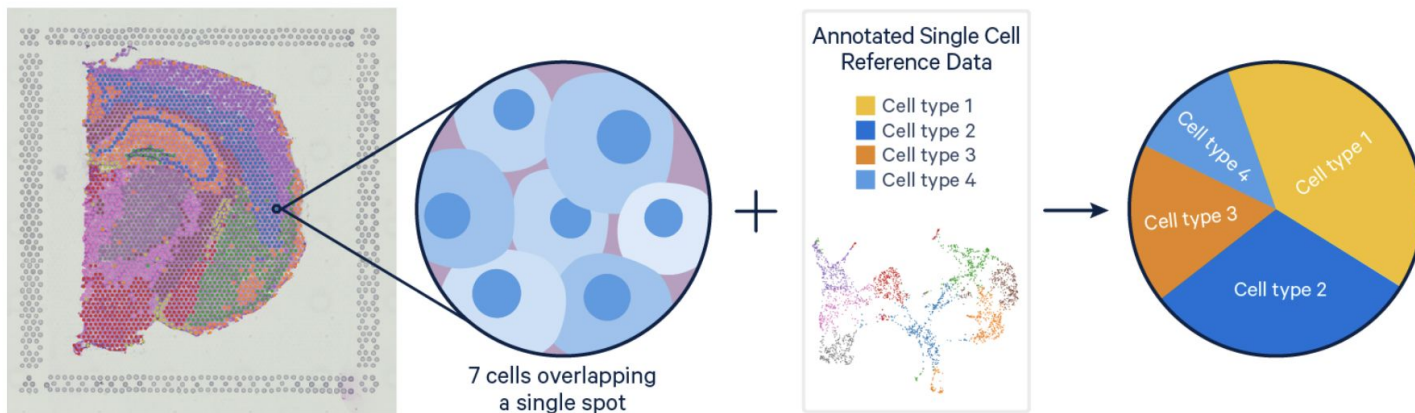
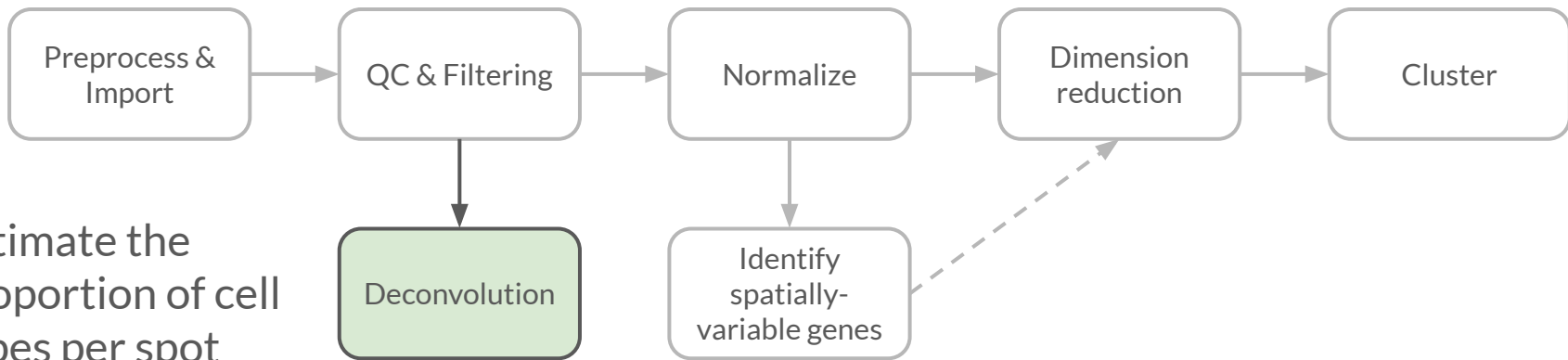
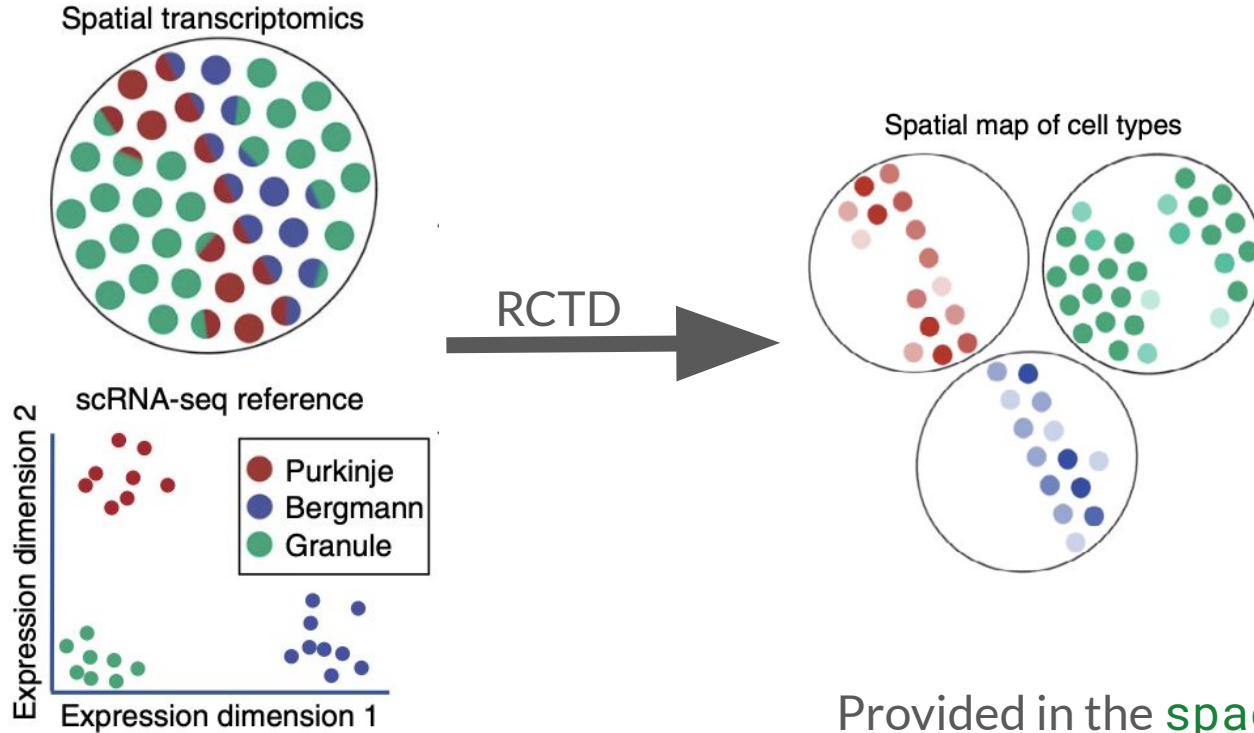


Performing deconvolution on spatial transcriptomics data

The Data Lab



We'll use RCTD: Robust Cell Type Deconvolution



Provided in the **spacexr** package

<https://doi.org/10.1038/s41587-021-00830-w>

Overview of the RCTD algorithm

1. Perform reference feature selection, times two:
 - a. Set 1: Get differentially expressed genes (DEGs) in the reference that distinguish cell types. Use for deconvolution itself.
 - b. Set 2: Get some less stringently-selected DEGs. Use for estimating platform effects.
2. Create deconvolution reference cell type profiles from Set 1 DEGs as the mean per-gene expression per cell type
3. Estimate global, fixed per-gene platform effects from the Set 2 of DEGs
4. Perform deconvolution using maximum likelihood
 - a. Feed cell type profiles & platform effects into model to infer proportions of each cell type per spot

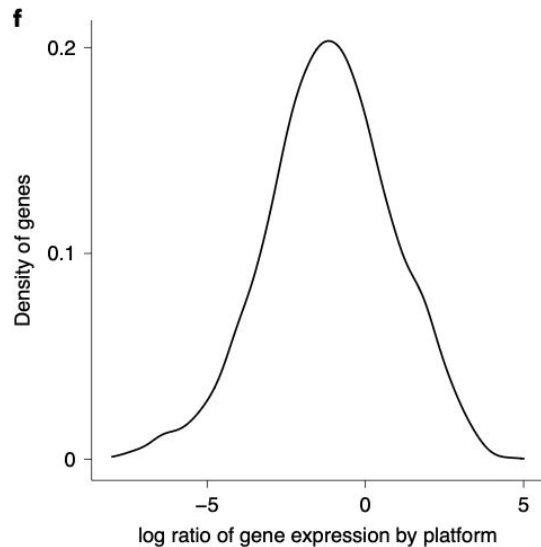


Why do we need a "platform effect"?

The reference is single-cell/nuclei, but the query data is Visium spot data

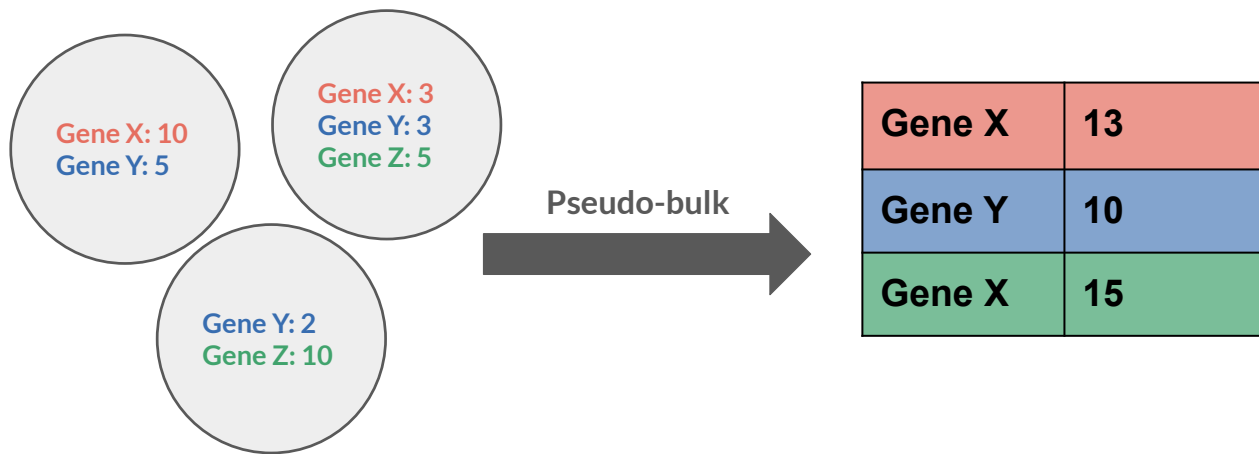
Different technologies have different gene capture rates, and we don't want this technical difference between the reference and Visium data to bias our results

Ratio of single-cell to single-nuclei gene expression from same tissue



How RCTD infers platform effects

1. Pseudo-bulk the spatial transcriptomics expression across all spots with the larger set of DEGs



2. Use this pseudo-bulked spatial expression to estimate gene-specific scaling parameters which are used to bring Visium/reference onto the same scale during deconvolution

The RCTD model has three modes

- Doublet mode
 - Here, doublet means “a spot with exactly two cells”
 - RCTD fits two constrained models: fixed to two cells, and fixed to one cell ("singlet") per spot
 - RCTD uses AIC model selection to determine if each spot is more likely a doublet or a singlet
- Multi-mode
 - You provide a maximum number of cell types possible per spot, and RCTD constrains the model to your maximum
- Full mode
 - RCTD fits an unconstrained model
 - Caution: Per-spot proportions are not guaranteed to sum to 1 under this mode, so you need to normalize each spot....probably.

Results depend on a close match between your data and your reference

- You can't detect cell types that aren't in your reference, even if they are present
- Commonly, papers will perform both scRNA-seq and spatial transcriptomics in order to have a well-matched reference



Interpreting deconvolution results

- Per spot, you get a proportion of each cell type. Are those proportions realistic?
 - Is a 15% proportion biologically realistic signal, or is it noise? It depends!
 - There are somewhere between 1-20 cells per spot — your biological knowledge about the tissue (or the H&E itself) may help pin that down
- Proportions are not absolute counts
 - How many cells is 90%? It depends!



On Visium data, deconvolution is mostly descriptive

- Explore the overall landscape of cellular composition in your tissue
- Identify tissue regions enriched for certain cell types (possibly "niches")
- Results may assist in labeling clusters: Are spots in certain clusters enriched for certain cell types?
- Results generally don't serve as input for downstream analyses
- Explicit cell-level analyses generally require higher resolution data
 - Neighborhood identification
 - Cell adjacency analysis
 - Cell-cell communication analyses
 - Rigorous characterization of the tumor microenvironment, e.g. if immune infiltration is rare

